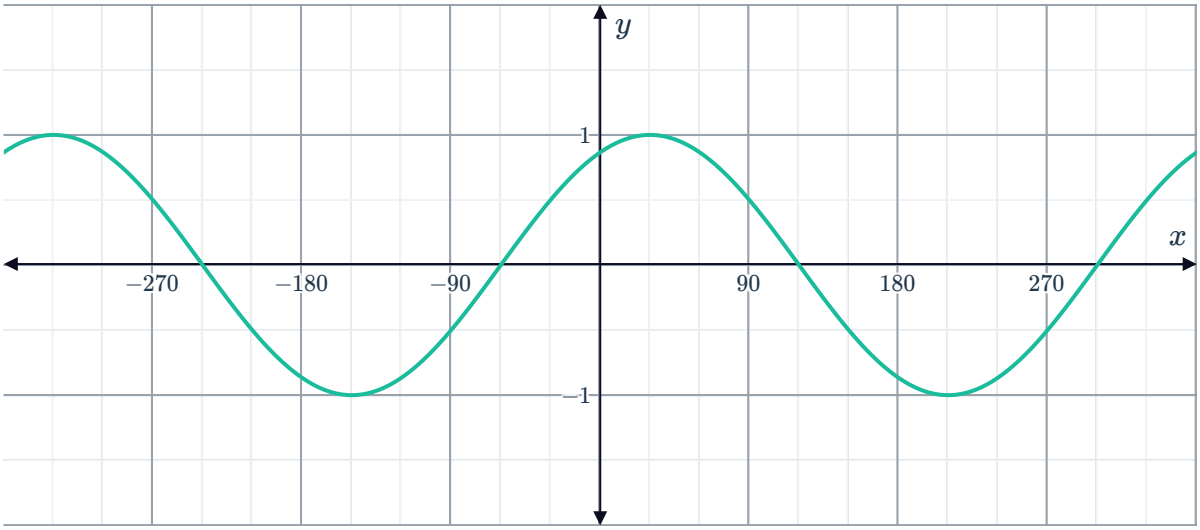
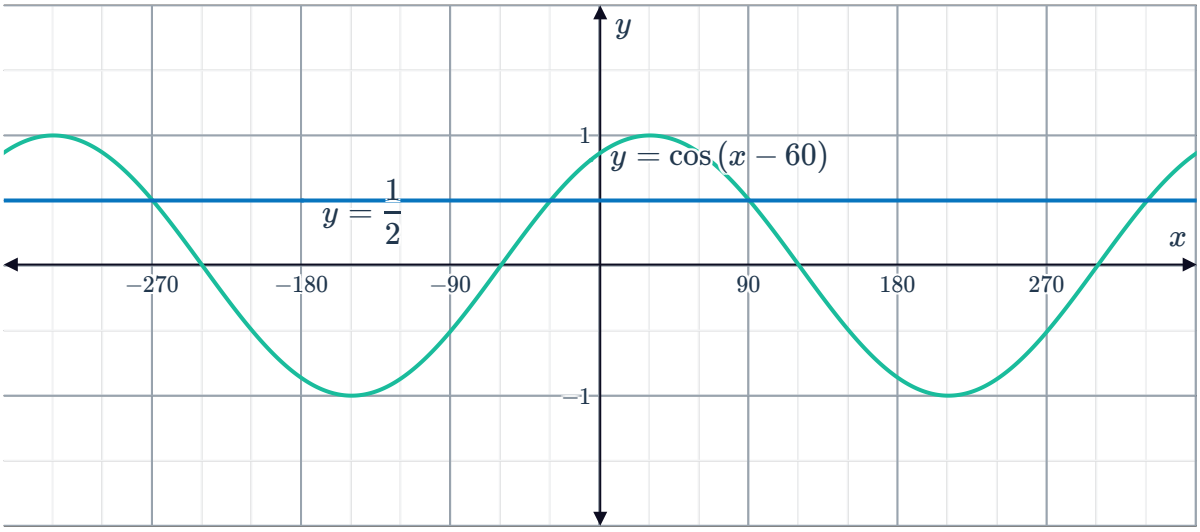
# Graphical solutions

**1**

**a**



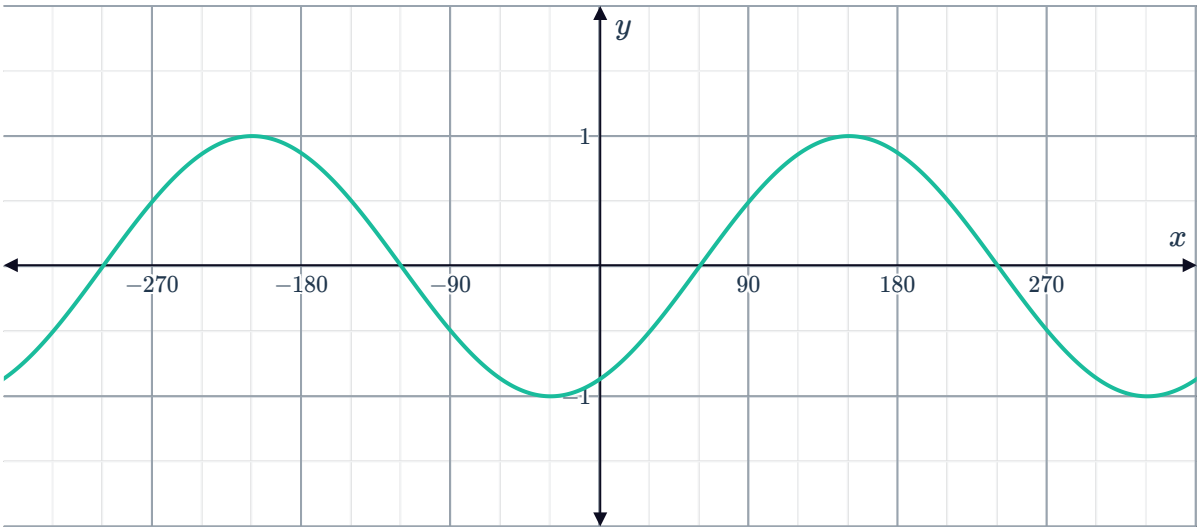

**b**




**c** $x = -270, -30, 90, 330$

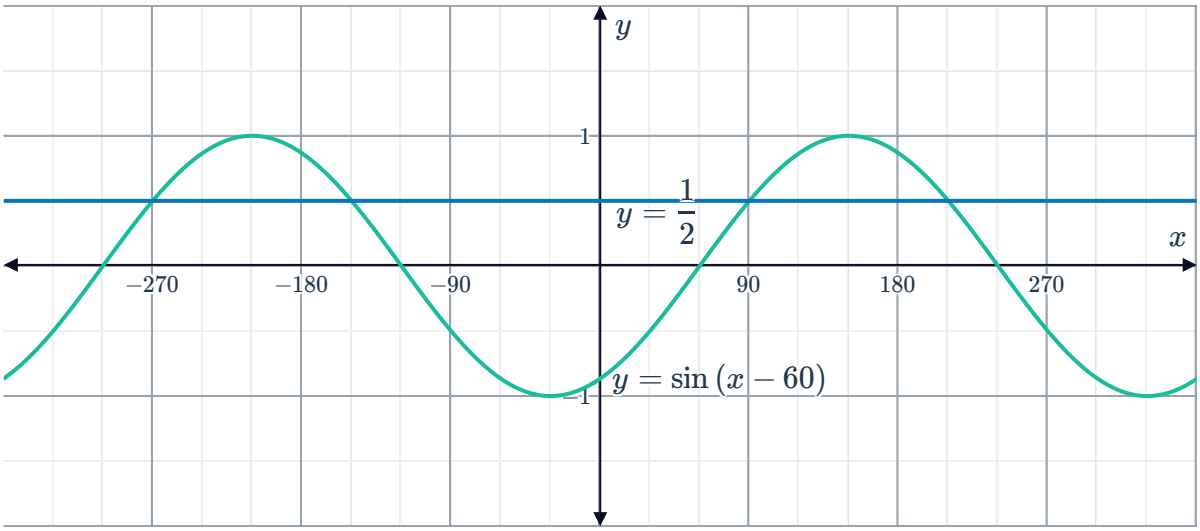
**2**

**a**




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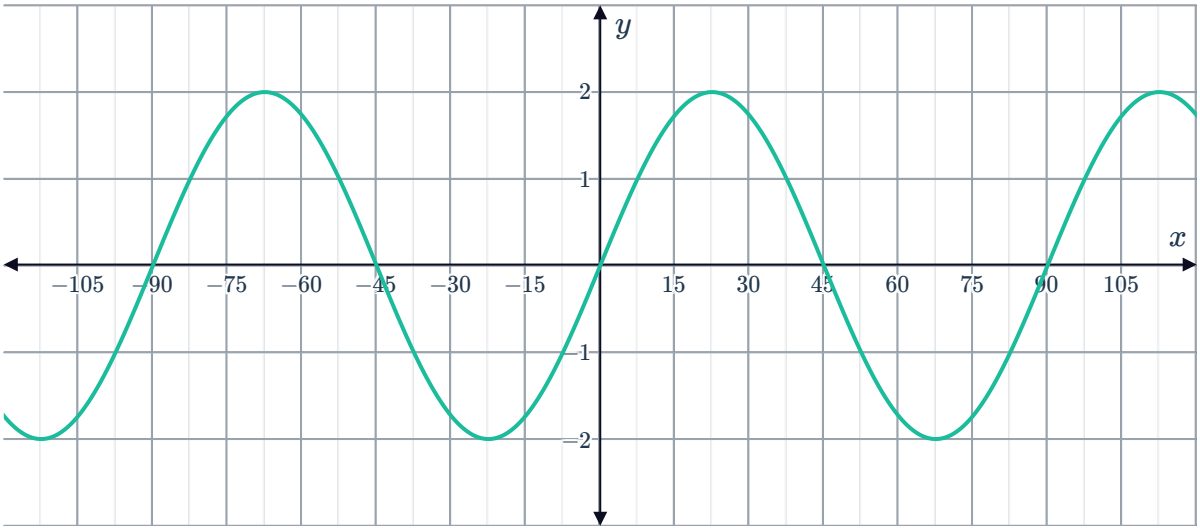
**b**



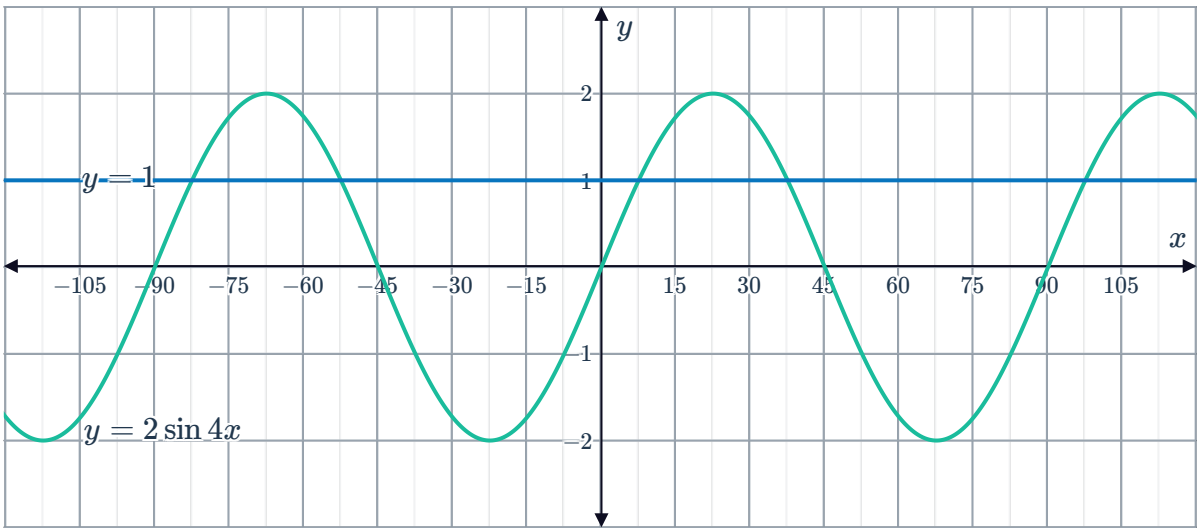

**c** $x = -270, -150, 90, 210$

**3**

**a**



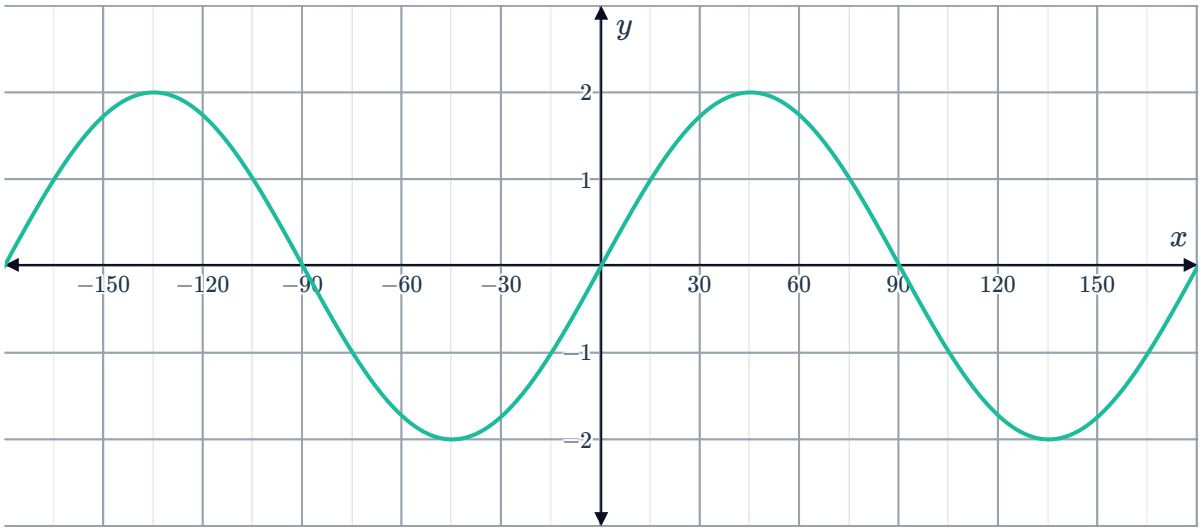

**b**




**c** $x = -82.5, -52.5, 7.5, 37.5$

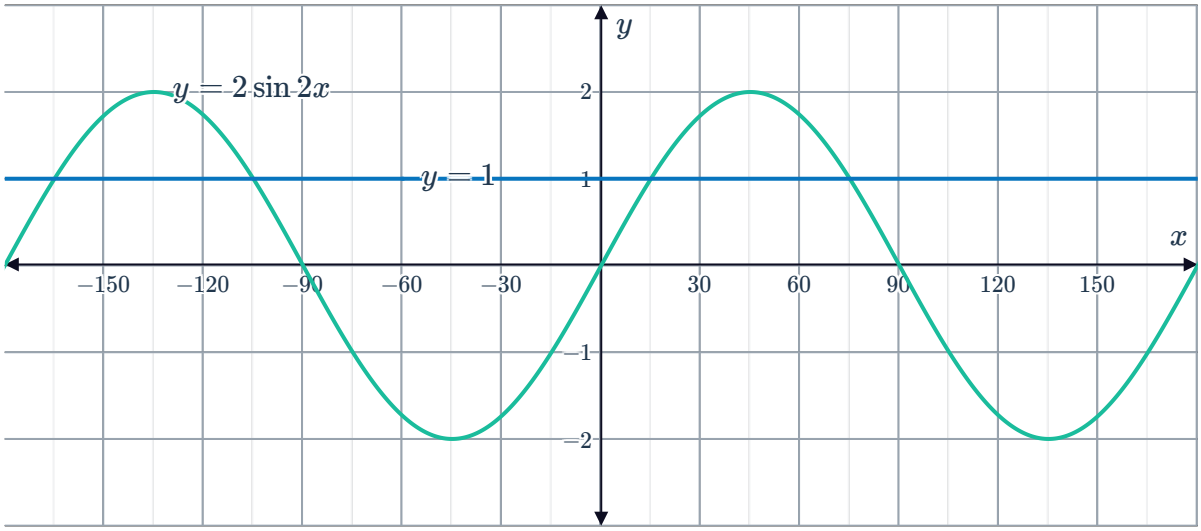
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**a**




**b** $y = 1$

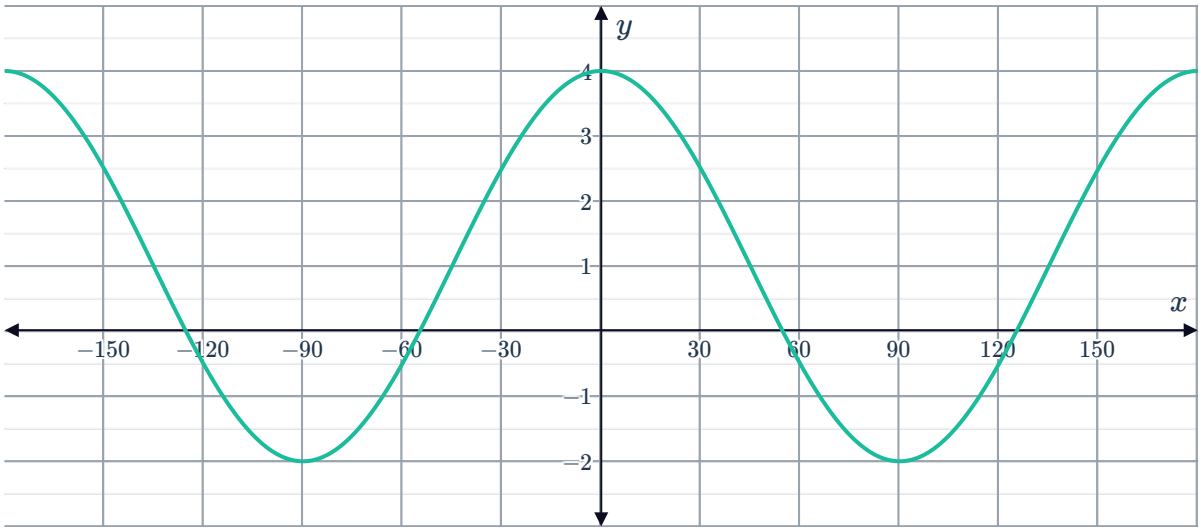
**c**




**d** $x = -165, -105, 15, 75$

**5**

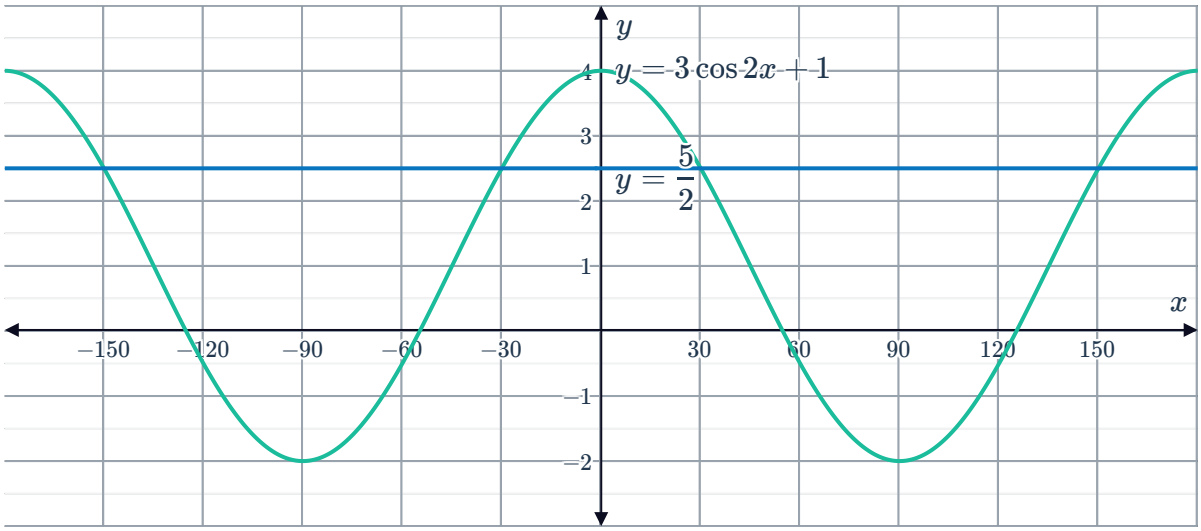
**a**




**b** $y = \frac{5}{2}$

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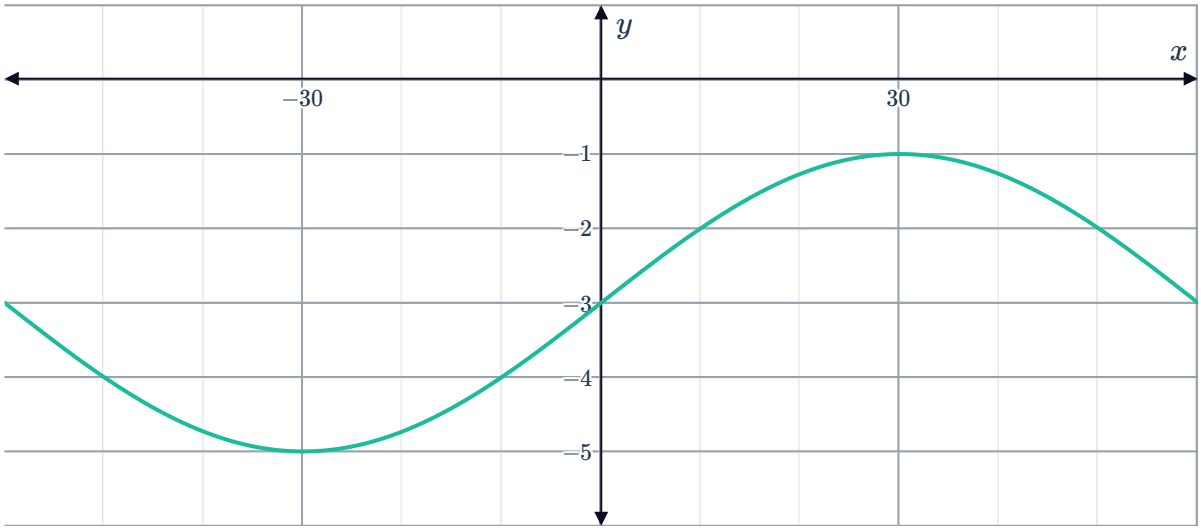
**c**




**d** $x = -150, -30, 30, 150$

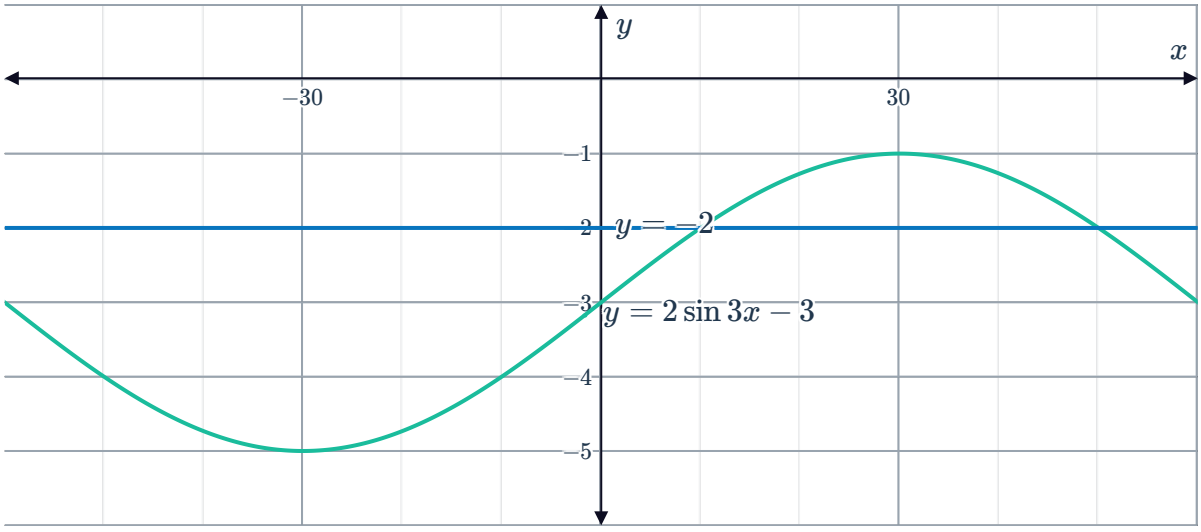
**6**

**a**




**b** $y = -2$

**c**

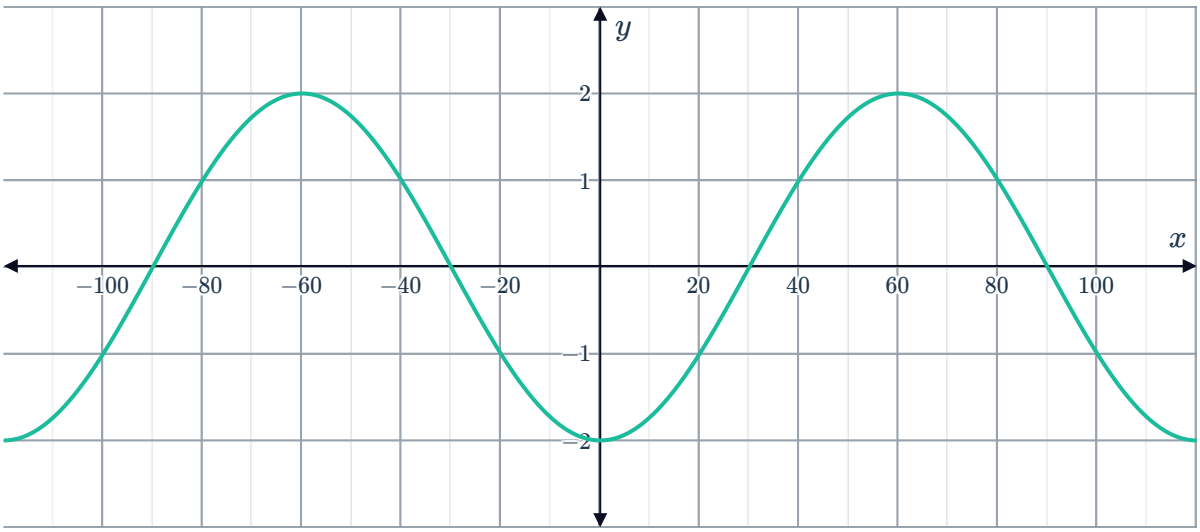



**d** $x = -110, -70, 10, 50$

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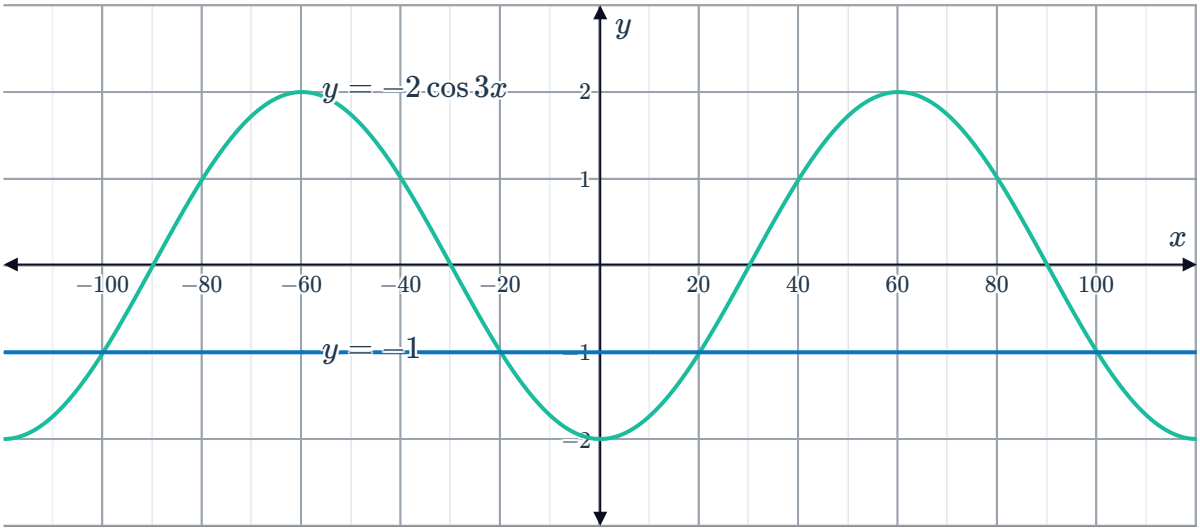
**7**

**a**




**b** $y = -1$

**c**




**d** $x = -100, -20, 20, 100$

## Technology

**8**

**a** $x = 12.9, 167.1, 192.9, 347.1$

**b** $x = 11.8, 78.2$

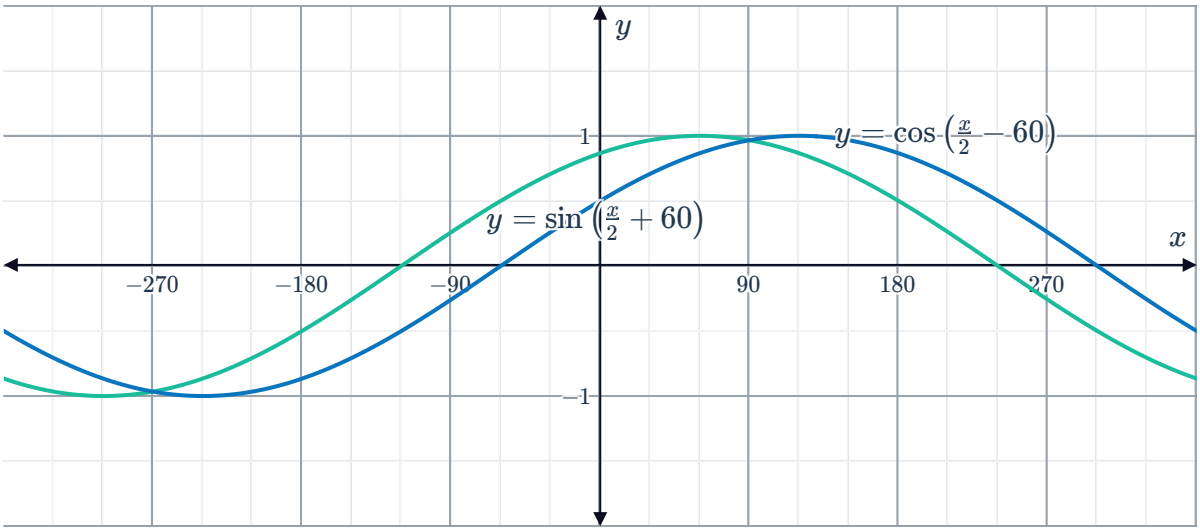
**c** $x = 509.7$

**9**

**a** $y = \sin\left(\frac{x}{2} + 60\right), y = \cos\left(\frac{x}{2} - 60\right)$

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**b**




**c** $x = -270, 90$

**10**

**a** $x = 0, 102, 180, 258$

**b** $x = 101, 303$

**c** $x = 35, 215$

**d** $x = 65, 142, 218, 295$

**e** $x = 7, 85, 180, 275, 353$

**f** $x = 29, 92, 147, 213, 268, 331$

**g** $x = 180, 211, 304$

**h** $x = 133, 227$

**11**

**a** $x = 101.18, 302.68$

**b** $x = 0, 18.23, 180, 284.62, 360.17$

**c** $x = -78.94$

**d** $x = 37.55, 115.05, 245.18, 322.62$

**e** $x = 34.85, 214.91$

**f** $x = 7.22, 84.73, 180, 275.45, 360$

**g** $x = 180, 210.96, 304.22$

**h** $x = 139.07, 221.10$

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